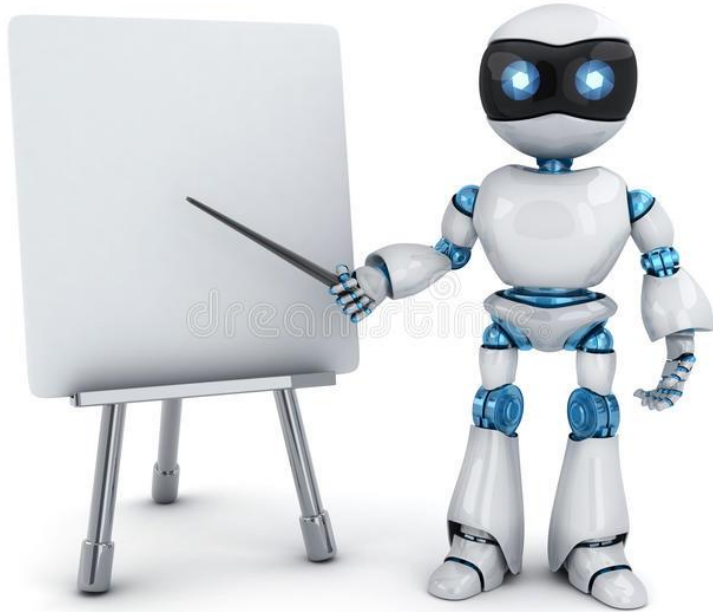
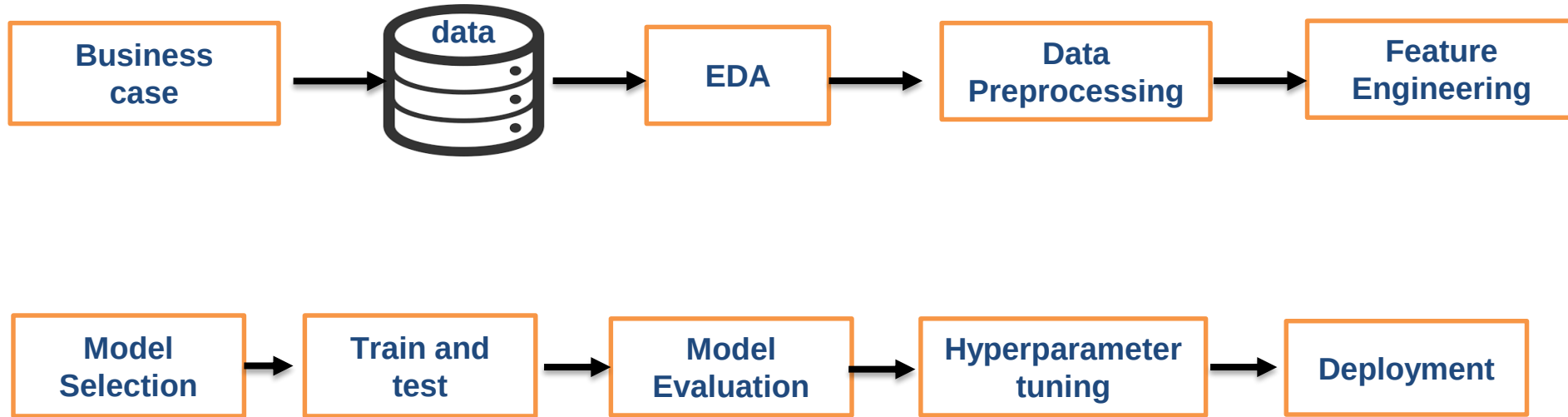




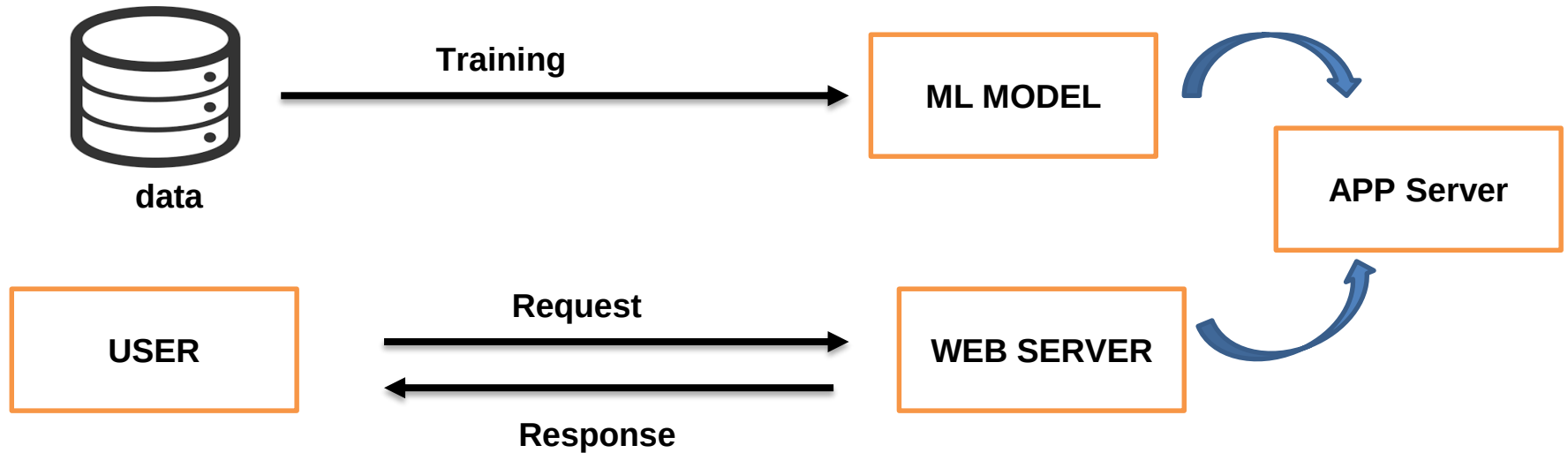
MODEL DEPLOYMENT

MACHINE LEARNING PIPELINE



What is model deployment?

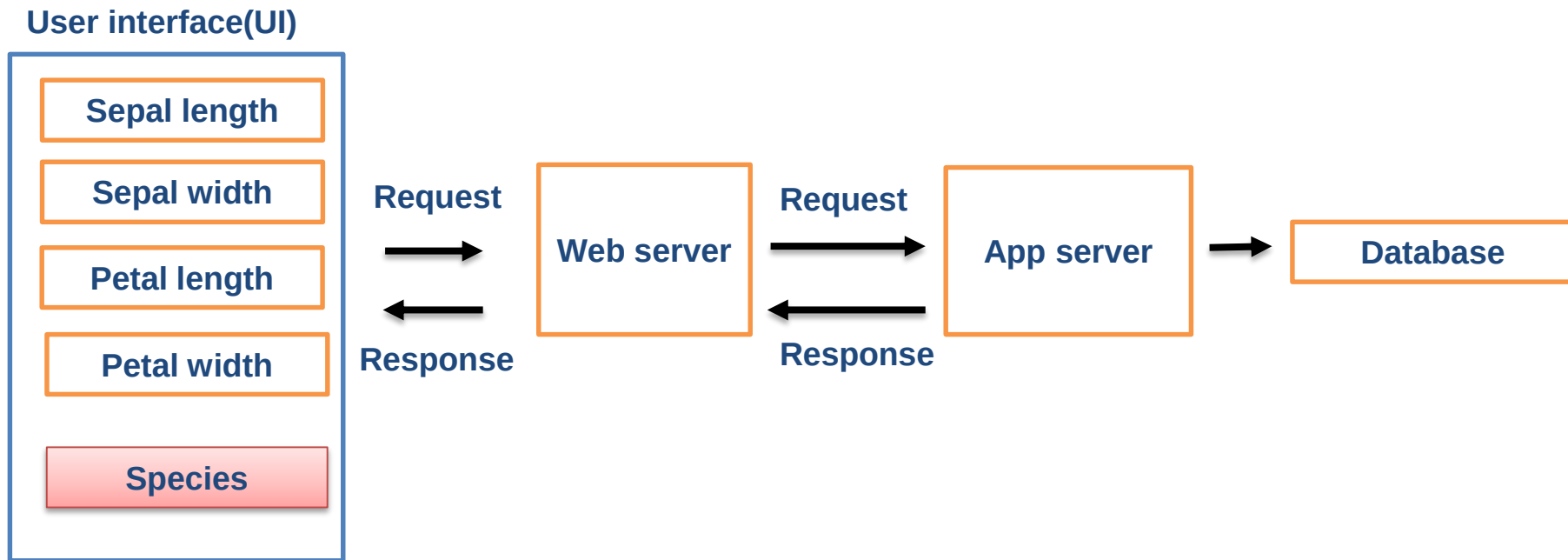
Model deployment is the process of putting machine learning model into production where it can take input from user and return an output.



WAYS TO DEPLOY MODEL



HOW DEPLOYMENT WORKS?



USER INTERFACE

User interface is the space where human can interact with computer , website or application.

WEBSERVER

A webserver is a software that receives request from clients/user and sends it to the application server.



APPLICATION SERVER

It is a program on computer that handles all application operations between users and organizations business applications or databases. It will process web server request and return a response.



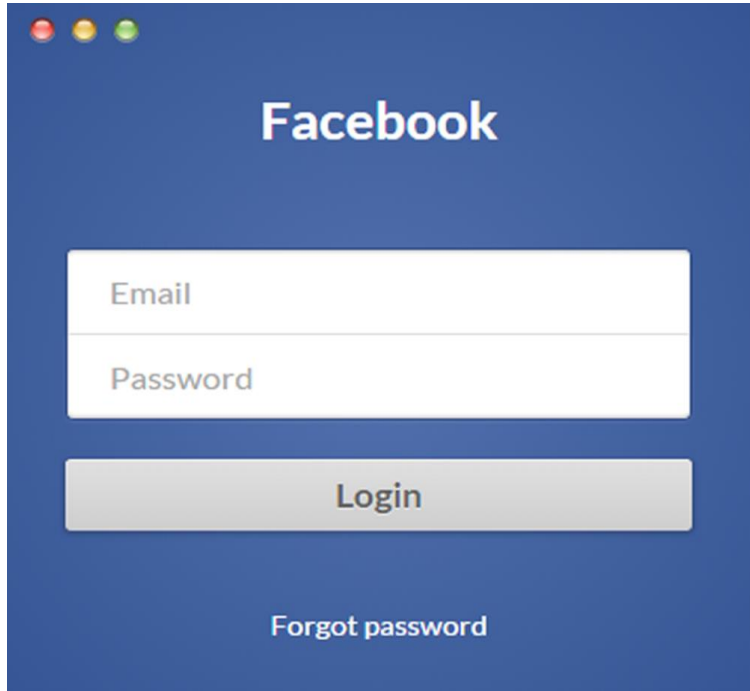
API

- API stands for Application programming interface.
- An API is set of rules that explain how computer or applications communicate with one another
- API is between an application and web server acting as intermediary layer that process data transfer between systems.

HOW API WORKS?

- Client application initiate an API call to retrieve information- also known as request.
- User requests are received by the API and that is sent to the web server.
- Server sends a response to the API with the requested information.
- The API transfers the data to the initial requesting application.

API Examples



A screenshot of the Facebook login interface. It features a dark blue background with the Facebook logo at the top. Below the logo are two white input fields for 'Email' and 'Password'. A large grey 'Login' button is positioned below the password field. At the bottom, there is a link for 'Forgot password'.

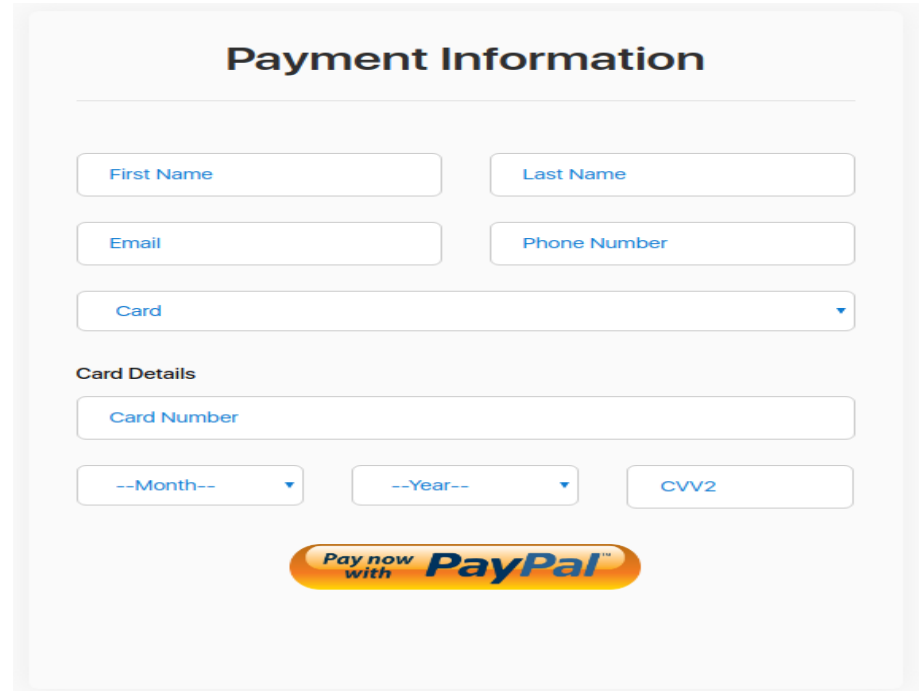
Facebook

Email

Password

Login

Forgot password



A screenshot of a 'Payment Information' form. The title 'Payment Information' is at the top. Below it are four input fields: 'First Name', 'Last Name', 'Email', and 'Phone Number'. A 'Card' dropdown menu is also present. Under the heading 'Card Details', there is a 'Card Number' field, a '--Month--' dropdown, a '--Year--' dropdown, and a 'CVV2' field. At the bottom, there is a 'Pay now with PayPal' button.

Payment Information

First Name

Last Name

Email

Phone Number

Card

Card Details

Card Number

--Month--

--Year--

CVV2

Pay now with PayPal

FRAME WORKS

- To create an API we need Frame works.
- Frameworks is a collection of libraries , methods and functions that provides a foundation for developing software applications.
- Different frameworks are **Flask API** , Django, Streamlit, web2py etc .

FLASK

- Flask is a open source web framework written in python that helps to develop web applications easily.
- It has two protocols

1) WSGI [Web server gateway interface]

Sending requests from web server to application server is done
By WSGI.



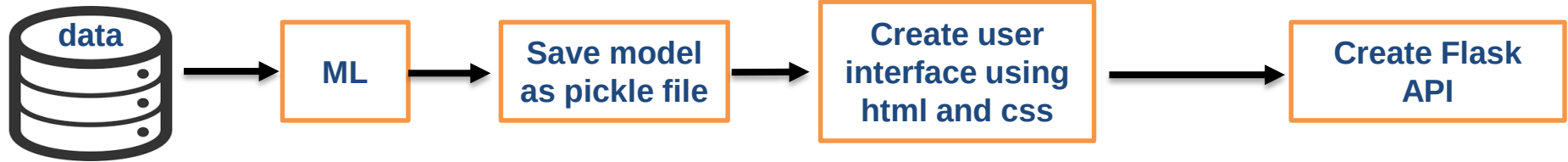
2) Jina 2 Template

It allows you to send the data from server side to user application when user makes the request.

DEPLOYMENT STEPS

- Create Model
- Save the model as pickle file
- Create user interface
- Using flask take inputs from user
- Using flask send the output

MACHINE LEARNING PIPELINE



Files required for model deployment are

- model.py
- model.pkl
- index.html
- main.py

Flask application

```
from flask import Flask

app = Flask(__name__)

@app.route('/')
def hello():
    return "Hello, Flask!"

if __name__ == '__main__':
    app.run(debug=True)
```

BASICS TERMS

POST

Used to receive the input.

ROUTE

It binds URL to the function.

PICKLE

It saves python object as a binary file.

HTML and CSS

HTML is a language used to create web pages and CSS is used for styling web pages.